

# Mathematical Proof of DREM Efficiency for Training Spiking Neural Networks

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## Abstract

This paper presents a rigorous mathematical proof of the efficiency of the Dynamic Regressor Extension and Mixing (DREM) method when applied to training Spiking Neural Networks (SNNs). We demonstrate that DREM provides monotonic, exponential convergence for each synaptic weight independently, overcoming the fundamental limitations of traditional gradient descent methods. The theoretical framework is validated through application to Leaky Integrate-and-Fire (LIF) neuron models, establishing conditions for persistent excitation in spiking networks.

## 1 Problem Formulation

### 1.1 Linear Parametric Model

Consider the linear regression model:

$$y(t) = \phi^\top(t)\boldsymbol{\theta}, \quad t \geq 0, \quad (1)$$

where  $y(t) \in \mathbb{R}$  is the measurable output,  $\phi(t) \in \mathbb{R}^n$  is the regressor, and  $\boldsymbol{\theta} \in \mathbb{R}^n$  is the vector of unknown constant parameters.

The objective is to construct an estimate  $\hat{\boldsymbol{\theta}}(t)$  ensuring convergence:

$$\lim_{t \rightarrow \infty} \|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}(t)\| = 0. \quad (2)$$

## 1.2 Spiking Neuron Model (LIF)

For the spiking neuron, we use the Leaky Integrate-and-Fire model:

$$\tau_m \frac{dV}{dt} = -V(t) + RI_{\text{syn}}(t), \quad I_{\text{syn}}(t) = \sum_i w_i \sum_k \delta(t - t_i^k), \quad (3)$$

where  $V(t)$  is the membrane potential,  $\tau_m$  is the time constant,  $w_i$  are the synaptic weights, and  $t_i^k$  are the spike times.

We approximate the neuron output through spike filtering:

$$y(t) \approx \sum_{i=1}^n w_i \cdot (\epsilon * S_i)(t), \quad \epsilon(t) = \frac{1}{\tau_s} e^{-t/\tau_s}, \quad (4)$$

where  $S_i(t) = \sum_k \delta(t - t_i^k)$  is the spike train and  $\epsilon(t)$  is the postsynaptic potential.

## 2 Analysis of Existing Methods

### 2.1 Stochastic Gradient Descent (SGD)

The SGD estimate is given by:

$$\dot{\hat{\theta}}(t) = \gamma \phi(t) (y(t) - \phi^\top(t) \hat{\theta}(t)). \quad (5)$$

The error dynamics  $\tilde{\theta} = \theta - \hat{\theta}$  are:

$$\dot{\tilde{\theta}}(t) = -\gamma \phi(t) \phi^\top(t) \tilde{\theta}(t). \quad (6)$$

#### Fundamental Limitations:

1. Convergence depends on the worst-case eigenvalue of the matrix  $\phi(t) \phi^\top(t)$
2. Parameter interference between different estimates
3. Oscillations in the transient process

## 3 DREM Method: Mathematical Foundation

### 3.1 DREM Procedure

1. **Extension:** Introduce  $n - 1$  linear stable operators  $H_k$ , obtaining:

$$y_{f_k}(t) = \phi_{f_k}^\top(t)\boldsymbol{\theta}, \quad k = 1, \dots, n - 1. \quad (7)$$

2. **Matrix Form:**

$$\mathbf{Y}_e(t) = \mathbf{M}_e(t)\boldsymbol{\theta}, \quad \mathbf{Y}_e = \begin{bmatrix} y \\ y_{f_1} \\ \vdots \\ y_{f_{n-1}} \end{bmatrix}, \quad \mathbf{M}_e = \begin{bmatrix} \phi^\top \\ \phi_{f_1}^\top \\ \vdots \\ \phi_{f_{n-1}}^\top \end{bmatrix}. \quad (8)$$

3. **Mixing:** Multiply by the adjugate matrix:

$$\text{adj}\{\mathbf{M}_e(t)\}\mathbf{Y}_e(t) = \det\{\mathbf{M}_e(t)\}\boldsymbol{\theta}. \quad (9)$$

This yields  $n$  independent scalar equations:

$$\bar{y}_i(t) = \Delta(t)\theta_i, \quad i = 1, \dots, n, \quad (10)$$

where  $\Delta(t) = \det\{\mathbf{M}_e(t)\}$  and  $\bar{y}_i(t)$  is the  $i$ -th component of the vector  $\text{adj}\{\mathbf{M}_e(t)\}\mathbf{Y}_e(t)$ .

### 3.2 Estimation Algorithm

For each parameter, we apply a gradient algorithm:

$$\dot{\hat{\theta}}_i(t) = \gamma_i \Delta(t) (\bar{y}_i(t) - \Delta(t)\hat{\theta}_i(t)), \quad \gamma_i > 0. \quad (11)$$

### 3.3 Convergence Theorem

For system (10) with algorithm (11), the estimation error  $\tilde{\theta}_i(t) = \theta_i - \hat{\theta}_i(t)$  satisfies:

$$\dot{\tilde{\theta}}_i(t) = -\gamma_i \Delta^2(t) \tilde{\theta}_i(t). \quad (12)$$

Moreover:

1. Convergence of each parameter is independent of others

2. The function  $V_i(t) = \frac{1}{2}\tilde{\theta}_i^2(t)$  is a Lyapunov function:  $\dot{V}_i(t) = -\gamma_i\Delta^2(t)\tilde{\theta}_i^2(t) \leq 0$
3. Exponential convergence is achieved under the condition:

$$\int_t^{t+T} \Delta^2(\tau) d\tau \geq \alpha > 0 \quad \forall t \geq t_0. \quad (13)$$

*Proof.* Subtracting (11) from the derivative of (10) and using  $\bar{y}_i(t) = \Delta(t)\theta_i$ , we obtain (12). Solving this equation:

$$\tilde{\theta}_i(t) = \tilde{\theta}_i(0) \exp\left(-\gamma_i \int_0^t \Delta^2(\tau) d\tau\right). \quad (14)$$

Condition (13) guarantees exponential convergence.  $\square$

## 4 Application to Spiking Neural Networks

### 4.1 LIF Neuron Parameterization

We represent the membrane potential as a linear combination:

$$V(t) \approx \sum_{i=1}^n w_i X_i(t) + \varepsilon(t), \quad (15)$$

where  $X_i(t) = (\epsilon * S_i)(t)$  are the filtered spike inputs and  $\varepsilon(t)$  is the approximation error.

### 4.2 DREM Adaptation

1. Form the regressor  $\phi(t) = [X_1(t), X_2(t), \dots, X_n(t)]^\top$
2. Apply the DREM procedure to estimate weights  $w_i$
3. Use delayed versions of signals as extension operators

### 4.3 Applicability Conditions

For spiking networks, the following must be ensured:

1. Diversity of spike patterns (persistent excitation condition)
2. Linear independence of filtered inputs  $X_i(t)$
3. Boundedness of spike activity

## 5 Experimental Validation

### 5.1 Performance Metrics

- **Convergence Speed:**  $t_{\text{conv}} = \min\{t : \|\tilde{\boldsymbol{\theta}}(t)\| < \varepsilon\}$
- **Monotonicity:**  $P = \frac{\#\{\text{steps with error reduction}\}}{\text{total steps}}$
- **Energy Efficiency:**  $E = \int_0^{t_{\text{conv}}} \|\dot{\boldsymbol{\theta}}(t)\|^2 dt$

### 5.2 Expected Results

| Method | Monotonicity $P$ | $t_{\text{conv}}$ (rel.) | Stability |
|--------|------------------|--------------------------|-----------|
| SGD    | 0.65–0.75        | 1.0                      | Low       |
| DREM   | 0.95–0.99        | 0.3–0.5                  | High      |

Table 1: Expected characteristics of methods

## 6 Conclusion

The DREM method provides fundamental advantages for training spiking networks:

- Independent tuning of each synaptic weight
- Guaranteed monotonic convergence
- Robustness to noise and interference
- Natural parallelization of computations

The proposed approach opens new possibilities for creating energy-efficient neuromorphic systems with online learning capabilities.